

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-II]

TEST # 04

DATE : 07-07-2019

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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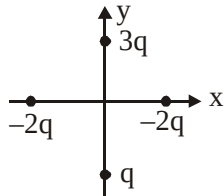
Your Target is to secure Good Rank in NEET 2020

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

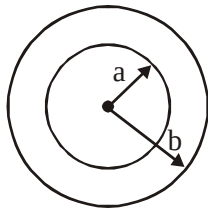
BEWARE OF NEGATIVE MARKING

1. 4 charges are placed each at a distance 'a' from origin. The dipole moment of configuration is:



- (1) $2qa \hat{j}$ (2) $3qa \hat{j}$
(3) $2aq [\hat{i} + \hat{j}]$ (4) None

2. If the electric potential of the inner metal sphere is 10 volt & that of the outer shell is 5 volt, then the potential at the centre will be :



- (1) 10 volt (2) 5 volt
(3) 15 volt (4) 0 volt

3. The change in potential energy when a body of mass m is raised to height nR from the earth's surface is (where R is radius of earth and g is acceleration due gravity at surface of earth) :-

- (1) $mgR \left(\frac{n}{n-1} \right)$ (2) mgR
(3) $mgR \left(\frac{n}{n+1} \right)$ (4) $mgR \left(\frac{n^2}{n^2+1} \right)$

4. A satellite of the earth is revolving in a circular orbit with a uniform speed v . If the gravitational force suddenly disappears, the satellite will

- (1) Continue to move with velocity v along the original orbit
(2) Move with a velocity v , tangentially to the original orbit
(3) Fall down with increasing velocity
(4) Ultimately come to rest somewhere on the original orbit

5. A particle is thrown vertically upwards from the surface of earth and it reaches to a maximum height equal to the radius of earth. The ratio of the velocity of projection to the escape velocity on surface of earth is :-

- (1) $\frac{1}{\sqrt{2}}$ (2) $\frac{1}{2}$
(3) $\frac{1}{4}$ (4) $\frac{1}{2\sqrt{2}}$

6. A satellite which is geostationary in a particular orbit is taken to another orbit. Its distance from the center of earth in the new orbit is 2 times that of the earlier orbit. The time period in the new orbit is :-

- (1) 48 hrs. (2) $48\sqrt{2}$ hrs.
(3) 24 hrs. (4) $24\sqrt{2}$ hrs.

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| <p>7. The minimum energy required to launch m kg satellite from earth's surface in a circular orbit at an altitude $2R$. where R is the radius of earth is</p> <p>(1) $\frac{5}{3}mgR$ (2) $\frac{4}{3}mgR$</p> <p>(3) $\frac{5}{6}mgR$ (4) $\frac{5}{4}mgR$</p> <p>8. The atmosphere is held to the earth by</p> <p>(1) Winds
(2) Gravity
(3) Clouds
(4) None of the above</p> <p>9. Two point masses having mass M and $4M$ are placed at distance equal to r. The gravitational potential at a point where gravitational field intensity zero is :-</p> <p>(1) $-\frac{9GM}{r}$ (2) $-\frac{2GM}{3r}$</p> <p>(3) $-\frac{3GM}{r}$ (4) $-\frac{6GM}{5r}$</p> <p>10. A particle is projected vertically up with velocity $V = \sqrt{\frac{4gR_e}{3}}$ from earth surface. The velocity of particle at height equal to half of maximum height reached by it if g in acceleration due to gravity at surface of earth :-</p> <p>(1) $\sqrt{\frac{gR_e}{2}}$ (2) $\sqrt{\frac{gR_e}{3}}$</p> <p>(3) $\sqrt{gR_e}$ (4) $\sqrt{\frac{2gR_e}{3}}$</p> | <p>11. If the distance between two masses is doubled, the gravitational attraction between them</p> <p>(1) Is doubled
(2) Becomes four times
(3) Is reduced to half
(4) Is reduced to a quarter</p> <p>12. Two particles of masses 1 kg and 2 kg are placed at a separation of 50 cm. Assuming that the only forces acting on the particles are their mutual gravitation, find the initial acceleration of the heavier particle.</p> <p>(1) $2.67 \times 10^{-10} \text{ m/sec}^2$
(2) $4 \times 10^{-10} \text{ m/sec}^2$
(3) $5 \times 10^{-10} \text{ m/sec}^2$
(4) $6 \times 10^{-10} \text{ m/sec}^2$</p> <p>13. Two sphere of mass m and M are situated in air and the gravitation force between them is F. The space around the masses is now filled with a liquid of specific gravity 3. The gravitation force will now be :-</p> <p>(1) F (2) $\frac{F}{3}$ (3) $\frac{F}{9}$ (4) $3F$</p> <p>14. Two particles of equal mass go around a circle of radius R under the action of their mutual gravitation. The speed of each particle :-</p> <p>(1) $V = \frac{1}{2R} \sqrt{\frac{1}{GM}}$ (2) $V = \sqrt{\frac{GM}{2R}}$</p> <p>(3) $V = \frac{1}{2} \sqrt{\frac{GM}{R}}$ (4) $V = \sqrt{\frac{4GM}{R}}$</p> |
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15. Four particles of masses $2m$, $4m$, $6m$ and $8m$ are kept in sequence at the corners of a square of side a . The magnitude of gravitational force acting on a particle of mass ' m ' placed at the center of the square will be :-

- (1) Zero (2) $\frac{4\sqrt{2} Gm^2}{a^2}$
(3) $\frac{8\sqrt{2} Gm^2}{a^2}$ (4) $\frac{24 Gm^2}{a^2}$

16. The depth at which the value of acceleration due to gravity becomes $\frac{1}{n}$ times the value at the surface is (R be the radius of the earth) :

- (1) $\frac{R}{n}$ (2) $\frac{R}{n^2}$
(3) $\frac{R(n-1)}{n}$ (4) $\frac{Rn}{n-1}$

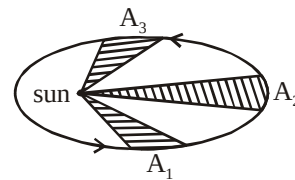
17. The earth is assumed to be a sphere of radius R . A platform is arranged at a height R from the surface of the earth. The escape velocity of a body from this platform is FV_e , where V_e is its escape velocity from the earth surface. The value of F is :

- (1) $\sqrt{2}$ (2) $\frac{1}{\sqrt{2}}$
(3) $\frac{1}{3}$ (4) $\frac{1}{2}$

18. A satellite of mass m orbiting closely around the earth in a circular orbit with a velocity v . What will be its energy ?

- (1) $\frac{3}{4}mv^2$ (2) $\frac{1}{2}mv^2$
(3) mv^2 (4) $-\frac{1}{2}mv^2$

19. A planet moving around sun sweeps area A_1 in 2 days, A_2 in 3 days and A_3 in 6 days. Then the relation between A_1 , A_2 and A_3 is :-



- (1) $3A_1 = 2A_2 = A_3$
(2) $2A_1 = 3A_2 = 6A_3$
(3) $3A_1 = 2A_2 = 6A_3$
(4) $6A_1 = 3A_2 = 2A_3$

20. Earth exerts a gravitational force on the Moon, keeping it in its orbit. The reaction to this force, in the sense of Newton's third law, is:

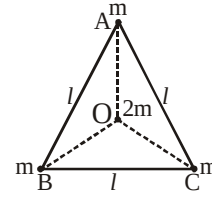
- (1) the centripetal force on the Moon
(2) the nearly circular orbit of the Moon
(3) the gravitational force on Earth by the Moon
(4) the tides due to the Moon

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21. The earth (mass $= 6 \times 10^{24} \text{ kg}$) revolves round the sun with angular velocity $2 \times 10^{-7} \text{ rad/s}$ in a circular orbit of radius $1.5 \times 10^8 \text{ km}$. The force exerted by the sun on the earth in *newtons*, is
- (1) 18×10^{25} (2) Zero
(3) 27×10^{39} (4) 36×10^{21}
22. The distance of the centres of moon and earth is D . The mass of earth is 81 times the mass of the moon. At what distance from the centre of the earth, the gravitational force will be zero
- (1) $\frac{D}{2}$ (2) $\frac{2D}{3}$
(3) $\frac{4D}{3}$ (4) $\frac{9D}{10}$
23. Reason of weightlessness in a satellite is
- (1) Zero gravity
(2) Centre of mass
(3) Zero reaction force by satellite surface
(4) None
24. Mass M is divided into two parts xM and $(1-x)M$. For a given separation, the value of x for which the gravitational attraction between the two pieces becomes maximum is
- (1) $\frac{1}{2}$ (2) $\frac{3}{5}$ (3) 1 (4) 2
25. A particle is dropped in a tunnel passing through centre of earth. The tunnel is very smooth. The distance of particle r is measured from earth's centre, then :-
- (1) acceleration of particle $\propto \frac{1}{r^2}$
(2) acceleration of particle $\propto \frac{1}{r}$
(3) acceleration of particle $\propto r$
(4) acceleration of particle is independent of r
26. Consider the following statements :
- S_1 : An object shall weigh more at pole than at equator when weighed by using a physical balance.
 S_2 : It shall weigh the same at pole and equator when weighed by using a physical balance.
 S_3 : It shall weigh the same at pole and equator when weighed by using a spring balance.
 S_4 : It shall weigh less at the pole than at equator when weighed using a spring balance.
- Which of the above statements is correct ?
- (1) S_1 (2) S_2 (3) S_3 (4) S_4
27. Satellites orbiting the earth have finite life and sometimes debris of satellites fall to the earth. This is because,
- (1) the solar cells and batteries in satellites run out.
(2) the laws of gravitation predict a trajectory spiralling inwards.
(3) of viscous forces changing the speed of satellite and hence height to gradually decrease.
(4) of collisions with other satellites.

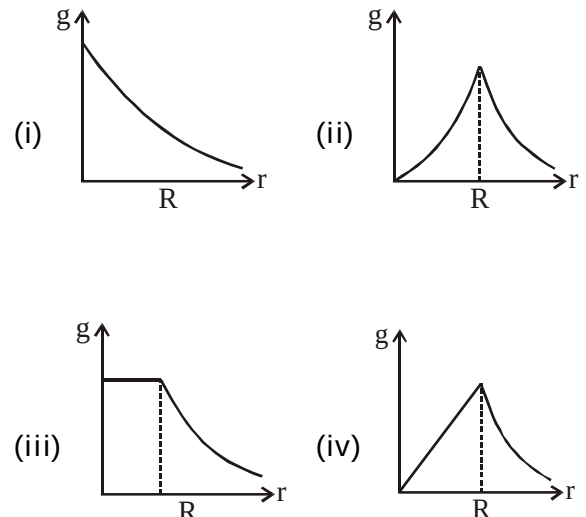
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28. If the speed of rotation of earth about its axis increases, then the apparent weight of the body at the equator will -
 (1) increase
 (2) decrease
 (3) remain unchanged
 (4) sometimes decrease and sometimes increase
29. Two spheres of same material of radius 'R' placed in such a way that they are touching to each other if the gravitation force between them is F then:-
 (1) $F \propto R^2$ (2) $F \propto R^4$
 (3) $F \propto \frac{1}{R^2}$ (4) $F \propto \frac{1}{R^4}$
30. If the universal constant of gravitation were decreasing uniformly with time, then a satellite in orbit would still maintain its :
 (1) Radius
 (2) Tangential speed
 (3) Angular momentum
 (4) Period of revolution
31. A body weighs 72 N on the surface of the earth. What is the gravitational force on it due to the earth at a height equal to half the radius of the earth?
 (1) 16 N (2) 28 N
 (3) 32 N (4) 72 N
32. Three masses each of mass m are placed at the vertices of an equilateral triangle ABC of side l as shown in figure. The force acting on a mass $2m$ placed at the centroid O of the triangle is :-



- (1) zero (2) $\frac{2Gm^2}{l^2}$
 (3) $\frac{4Gm^2}{l^2}$ (4) $\frac{6Gm^2}{l^2}$

33. The dependence of acceleration due to gravity g on the distance r from the centre of the earth assumed to be a sphere of radius R of uniform density is as shown figure below :-



The correct figure is

- (1) (i) (2) (ii)
 (3) (iii) (4) (iv)

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34. The weight of an object in the coal mine, sea level, at the top of the mountain are w_1, w_2 and w_3 respectively, then
- $w_1 < w_2 > w_3$
 - $w_1 = w_2 = w_3$
 - $w_1 < w_2 < w_3$
 - $w_1 > w_2 > w_3$
35. Two identical spheres each of mass M and radius R are separated by a centre to centre distance $10R$. The gravitational force on mass m placed at the midpoint of the line joining the centres of the spheres is :-
- zero
 - $\frac{2GMm}{25R^2}$
 - $\frac{GMm}{25R^2}$
 - $\frac{GMm}{100R^2}$
36. Dimensions of gravitational constant are :-
- $[ML^2T^2]$
 - $[M^1L^3T^{-2}]$
 - $[M^0L^3T^2]$
 - $[M^{-1}L^3T^{-2}]$
37. Which of the following statements about the gravitational constant is true
- It is a force
 - It has no unit
 - It depends on the value of the masses
 - It does not depend on the nature of the medium in which the bodies are kept.
38. In order to find time, the astronaut orbiting in an earth satellite should use
- A pendulum clock
 - A watch having main spring to keep it going
 - Either a pendulum clock or a watch
 - Neither a pendulum clock nor a watch
39. What is the energy required to place a satellite of mass m from orbit of radius $2R$ to the orbit of radius $4R$ around planet earth. (take mass of earth = M)
- $\frac{GMm}{8R}$
 - $\frac{GMm}{2R}$
 - $\frac{GMm}{6R}$
 - $\frac{GMm}{16R}$
40. R is the radius of the earth and ω is its angular velocity and g_p is the value of g at the poles. The effective value of g at the latitude $\lambda = 60^\circ$ will be equal to
- $g_p - \frac{1}{4}R\omega^2$
 - $g_p - \frac{3}{4}R\omega^2$
 - $g_p - R\omega^2$
 - $g_p + \frac{1}{4}R\omega^2$
41. The height of the point vertically above the earth's surface, at which acceleration due to gravity becomes 1% of its value at the surface is (Radius of the earth = R)
- $8R$
 - $9R$
 - $10R$
 - $20R$

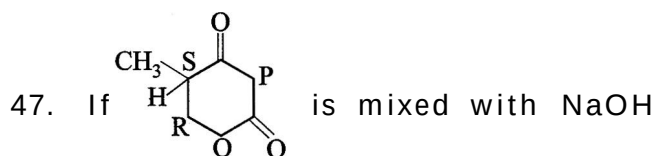
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42. What will be the acceleration due to gravity at height h if $h \gg R$. Where R is radius of earth and g is acceleration due to gravity on the surface of earth
- (1) $\frac{g}{\left(1 + \frac{h}{R}\right)^2}$
- (2) $g\left(1 - \frac{2h}{R}\right)$
- (3) $\frac{g}{\left(1 - \frac{h}{R}\right)^2}$
- (4) $g\left(1 - \frac{h}{R}\right)$
43. Escape velocity on earth is 11.2 km/s . What would be the escape velocity on a planet whose mass is 1000 times and radius is 10 times that of earth
- (1) 112 km/s
- (2) 11.2 km/s
- (3) 1.12 km/s
- (4) 3.7 km/s
44. Select the correct statement from the following
- (1) The orbital velocity of a satellite increases with the radius of the orbit
- (2) Escape velocity of a particle from the surface of the earth depends on the speed with which it is fired
- (3) The time period of a satellite does not depend on the radius of the orbit
- (4) The orbital velocity is inversely proportional to the square root of the radius of the orbit
45. In the absence of other conductors, the surface charge density
- (1) Is proportional to the charge on the conductor and its surface area
- (2) Inversely proportional to the charge and directly proportional to the surface area
- (3) Directly proportional to the charge and inversely proportional to the surface area
- (4) Inversely proportional to the charge and the surface area

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46. Inductive effect involves:

- (1) Delocalisation of α -electrons
- (2) Partial displacement of α -electrons
- (3) Delocalisation of π -electrons
- (4) Displacement of lone pair electrons.



solution, acid-base reaction occurs and snatches proton from organic molecule. Which carbon will lose proton easily?

- (1) P (2) Q
 - (3) R (4) S
48. The inductive effects of the group $-\text{CH}_3$, $-\text{COO}^-$, $-\text{Br}$, $-\text{NH}_3^+$ respectively are
- (1) +I, -I, +I, +I (2) -I, +I, -I, +I
 - (3) -I, -I, +I, +I (4) +I, +I, -I, -I
49. What is the correct increasing order of extent of deprotonation of the following acids in its aqueous solution?
- (I) $(\text{CH}_3)_2\text{CHCOOH}$
 - (II) $\text{CH}_3\text{CH}_2\text{COOH}$
 - (III) $\text{Cl}_2\text{CHCH}_2\text{COOH}$
 - (IV) $\text{CH}_3\text{CCl}_2\text{COOH}$
- (1) II < I < III < IV
 - (2) IV < III < I < II
 - (3) I < II < III < IV
 - (4) I < IV < III < II
50. Decreasing -I effect of given groups is :-
- (i) CN (ii) NO_2
 - (iii) NH_2 (iv) F

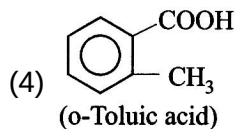
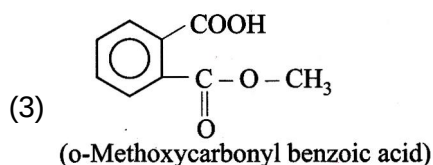
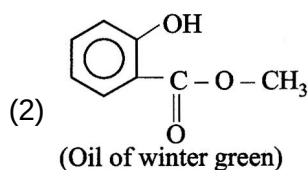
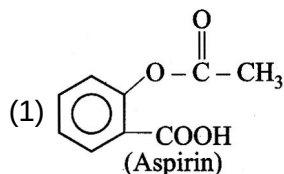
(1) iii > ii > i > iv

(2) ii > iii > iv > i

(3) iii > ii > iv > i

(4) ii > i > iv > iii

51. Find the strongest acid among the following compounds.



52. Which of the following alkyl groups has the maximum +I effect?

(1) $(\text{CH}_3)_2\text{CH}^-$

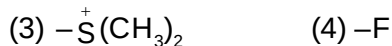
(2) $(\text{CH}_3)_3\text{C}^-$

(3) CH_3CH_2^-

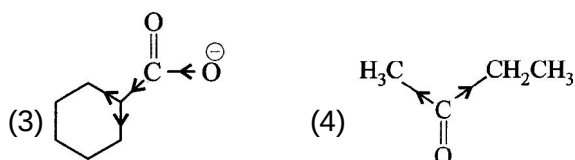
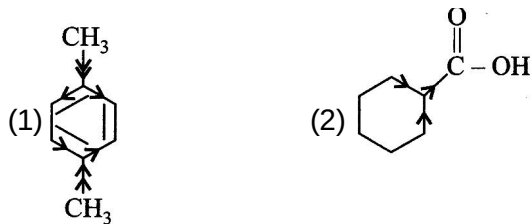
(4) CH_3^-

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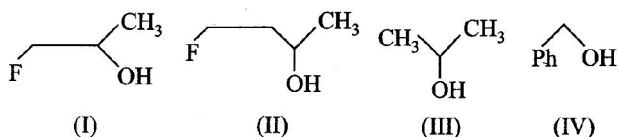
53. Which of the following is the strongest –I group?



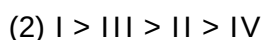
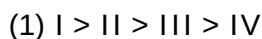
54. In which of the following species, incorrect direction of Inductive effect is shown?



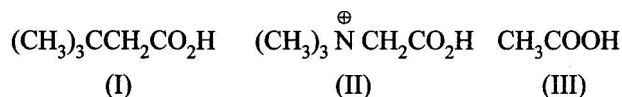
55. The order of reactivity of the following alcohols



towards conc. HCl is:



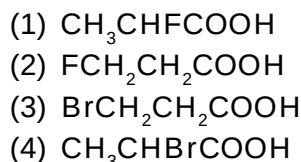
56. For the following acids



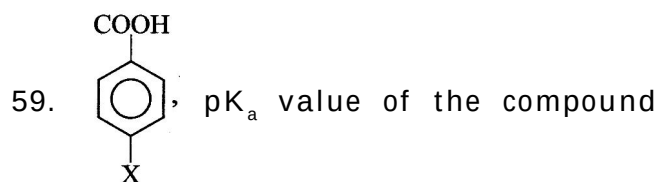
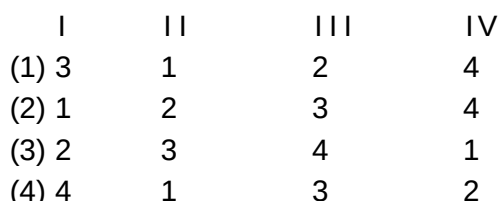
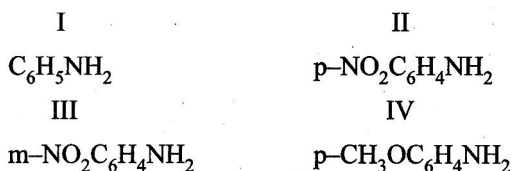
pK_a value will be in order



57. Which of the following acids has the smallest dissociation constant?



58. Assign number 1 for least to 4 for most to indicate the relative base strength of the following

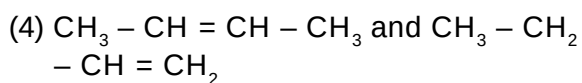
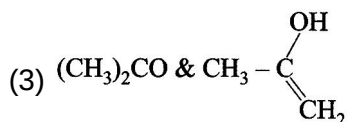
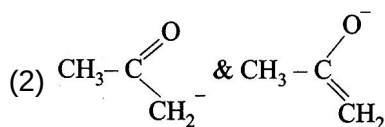
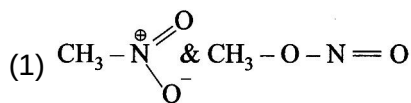


decreases if X is:



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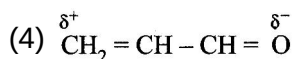
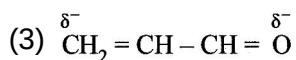
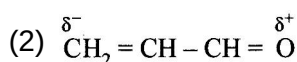
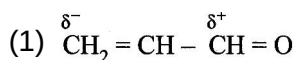
60. Which of the following pairs are resonating structures?



61. Which is the incorrect statement regarding resonance contributing structures?

- (1) All atoms in a contributing structures must have complete valence shells
- (2) All contributing structures must have same arrangement of atoms
- (3) All contributing structures must have same number of valence electrons
- (4) Equivalent resonance structures have same potential energies

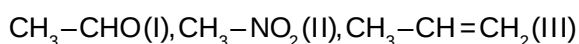
62. Polarization of electrons in acrolein may be written as:



63. The bond order of individual carbon-carbon bonds in benzene is

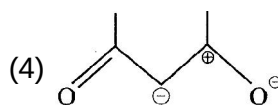
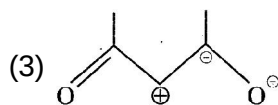
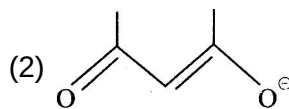
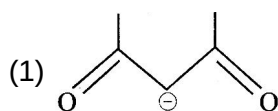
- (1) one
- (2) two
- (3) between one and two
- (4) one and two, alternately

64. What is the decreasing order of acidic strength of the following compounds for their most acidic hydrogen?



- (1) I > II > III
- (2) III > II > I
- (3) III > I > II
- (4) II > I > III

65. Which is the most stable resonating structure?

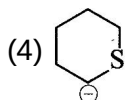
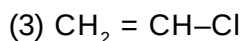
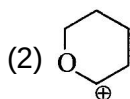
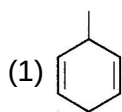


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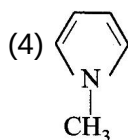
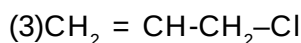
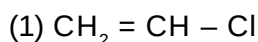
- | | |
|--|---|
| <p>66. Resonance effect involves:</p> <ol style="list-style-type: none"> (1) Delocalization of n-electrons along a conjugated system. (2) Delocalization of lone pair along a conjugated system. (3) Delocalization of negative charge along a conjugated system. (4) All are correct. <p>67. During derealization, which statement is incorrect :</p> <ol style="list-style-type: none"> (1) Net charge remains same (2) Number of paired electrons remain same (3) Number of unpaired electrons remain same (4) Energy of resonating structures always remains same <p>68. Which of the following resonating structures of 1-methoxy-1, 3-butadiene is least stable?</p> <ol style="list-style-type: none"> (1) $\ominus \text{CH}_2 - \text{CH} = \text{CH} - \text{CH} = \overset{\oplus}{\text{O}} - \text{CH}_3$ (2) $\text{CH}_2 = \text{CH} - \overset{\ominus}{\text{C}}\text{H} - \text{CH} = \overset{\oplus}{\text{O}} - \text{CH}_3$ (3) $\overset{\ominus}{\text{C}}\text{H}_2 - \overset{\oplus}{\text{C}}\text{H} - \text{CH} = \text{CH} - \text{O} - \text{CH}_3$ (4) $\text{CH}_2 = \text{CH} - \overset{\ominus}{\text{C}}\text{H} - \overset{\oplus}{\text{C}}\text{H} - \text{O} - \text{CH}_3$ <p>69. KOH can be used as drying agent for</p> <ol style="list-style-type: none"> (1) carboxylic acids (2) phenols (3) amines (4) esters | <p>70. Which of the following statements is incorrect?</p> <ol style="list-style-type: none"> (1) The energy of resonance hybrid is always less than that of any resonating structure. (2) The resonance energy is the difference between the enthalpies of formation of the molecule and the resonating structure having maximum energy. (3) The resonance structures are hypothetical structure and they do not represent any real molecule. (4) In delocalized structure of benzene the π-charge cloud is spread equally above and below the plane of molecule. <p>71. Which of the following statement regarding resonance is NOT correct?</p> <ol style="list-style-type: none"> (1) the different resonating structures of a molecule have fixed arrangement of atomic nuclei. (2) the different resonating structures differ in the arrangement of electrons. (3) the hybrid structure has equal contribution from all the resonating structures always. (4) None of the individual resonating structure explains all characteristics of the molecule. <p>72. Resonance energy is:</p> <ol style="list-style-type: none"> (1) The potential energy difference between most stable resonating structure and resonance hybrid (2) The potential energy difference between the least stable resonating structure and resonance hybrid (3) The potential energy difference between the least stable and most stable resonating structure (4) None |
|--|---|

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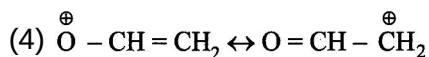
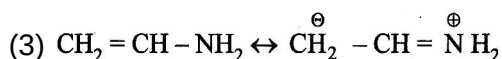
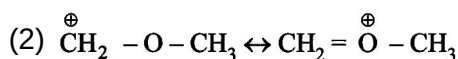
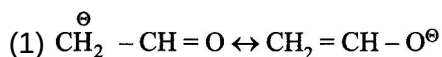
73. Resonance is not possible in:



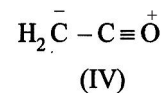
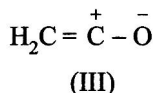
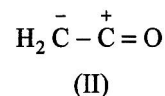
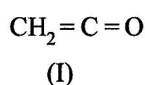
74. The compound which is not resonance stabilised



75. In which of the following first resonating structure is more stable than the second?



76. The correct stability order of the following resonance structures is:



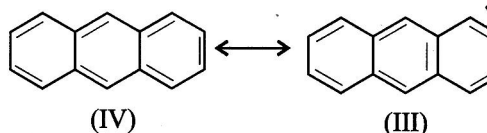
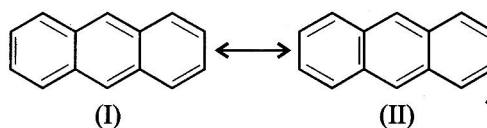
(1) (IV) > (I) > (III) > (II)

(2) (II) > (IV) > (I) > (III)

(3) (III) > (II) > (IV) > (I)

(4) (I) > (IV) > (III) > (II)

77. The correct stability order of given resonating structures is:



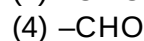
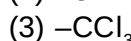
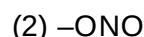
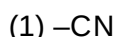
(1) I > II > III > IV

(2) IV > III > II > I

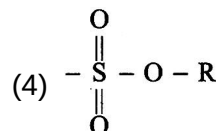
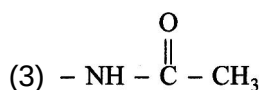
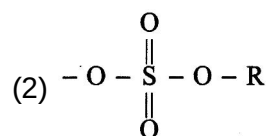
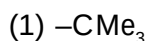
(3) I = II = III = IV

(4) II = III > I = IV

78. Which of the following group show +m effect?



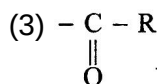
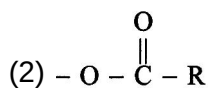
79. Which of the following group show -m effect?



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80. Which of the following group show +m > -I effect?

(1) -F



(4) -COOH

81. Which of the following group show -m and -I effect?

(1) -NO₂

(2) -NH₂

(3) -OH

(4) -Cl

82. Identify which of the following group acts as +m as well as -m?

(1) -F

(2) -NH₂

(3) -Cl

(4) -OH

83. +m and +I both effects are shown by:

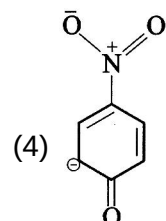
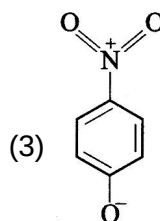
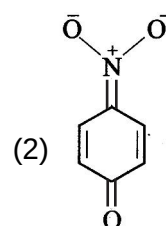
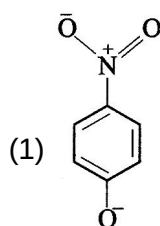
(1) - $\ddot{\text{O}}\text{H}$

(2) - $\ddot{\text{N}}\text{HCH}_3$

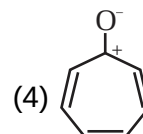
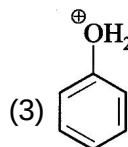
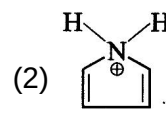
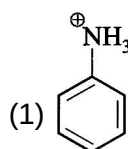
(3) -O[⊖]

(4) -C(CH₃)₃

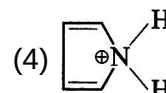
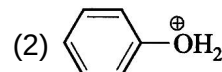
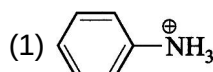
84. The most unlikely representation of resonance structures of p-nitrophenoxide ion is:



85. In which delocalisation of positive charge is possible?

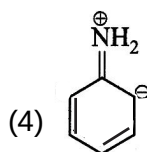
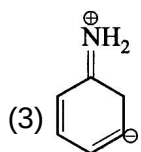
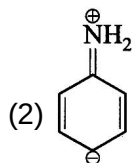
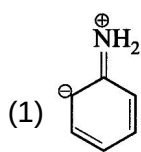


86. Resonance stabilized cation is:

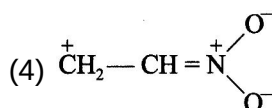
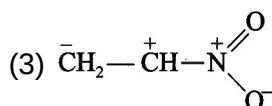
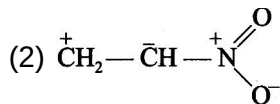
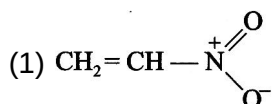


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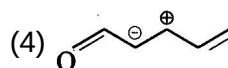
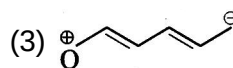
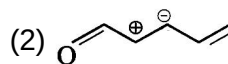
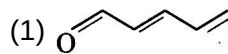
87. Which of the following is not the correct resonating structure of $C_6H_5NH_2$?



88. Least contributing resonating structure of nitroethene is:



89. The least and most stable resonating structures respectively are:



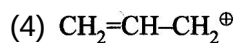
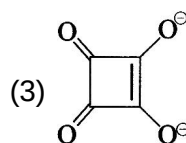
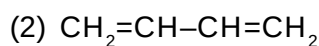
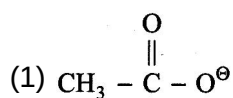
(1) 2, 1

(2) 2, 3

(3) 3, 1

(4) 1, 2

90. Which of the following species does not have equally contributing resonating structures?



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91. Find out the correct combination :-

	9 : 3 : 4	9 : 7	12 : 3 : 1
(1)	Complementary gene	Supplementary gene	Recessive epistasis
(2)	Supplementary gene	Complementary gene	Dominant epistasis
(3)	Recessive epistasis	Supplementary gene	Duplicate gene
(4)	Dominant epistasis	Recessive epistasis	Duplicate gene

92. Other than pea plants it was found that sometimes the F_1 had a phenotype that did not resemble either of the two parents and was in between the two. Due to :-

- (1) Complete dominance
- (2) Incomplete dominance
- (3) Co-dominance
- (4) Complementary gene interaction

93. If dominant C and P genes are essential for the development of purple colour in *Lathyrus odoratus* flowers. What would be the ratio of purple and white colour in a cross between $CcPp \times ccPp$.

- (1) 3 : 5
- (2) 9 : 7
- (3) 2 : 6
- (4) 4 : 4

94. Which of the following is a good example of multiple allele :-

- (1) ABO blood group
- (2) Size of starch grain in pea
- (3) Shape of seed
- (4) Flower colour in pea

95. Complementary genes were demonstrated by Bateson and Punnet in

- (1) *Capsella*
- (2) *Lathyrus odoratus*
- (3) *Antirrhinum*
- (4) *Mirabilis*

96. A woman (whose father is colour blind but mother normal) marries with haemophilic man with hypertrichosis. What percentage of progeny will show genotypically any two of the traits out of the three mentioned above at a given time :-

- (1) 0%
- (2) 25%
- (3) 50%
- (4) 75%

97. Two genes A and B present on same chromosome show 40% crossing over, then what will be the percentage of gamete (aB) for the plant having genotype AaBb (cis arrangement) :-

- (1) 40%
- (2) 20%
- (3) 80%
- (4) 10%

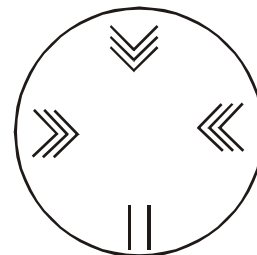
98. Red eye female *Drosophila* is crossed with white eye male. If this cross result is 50% progeny being red eye, we can say that :-

- (1) Both parents were hemizygous
- (2) female parent was heterozygous
- (3) female parent was pure
- (4) Both parents were pure

99. The leaf colour of a certain plant is controlled by one gene. In this gene, the allele G produces orange colour and g produces green colour. You have a plant with orange leaves, but you do not know its genotype. If you cross your unknown genotype plant with one of the plant whose genotype is listed below, you will be able to determine your plant genotype, with which plant should you cross it ?

- (1) GG
- (2) Gg
- (3) gg
- (4) All

100. Karyotype of drosophila given below then find out sex ratio of drosophila



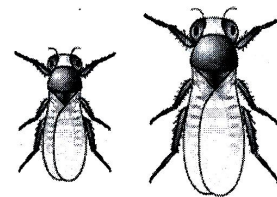
- (1) 2.5, supermale
- (2) 1.5, supermale
- (3) 0.66, intersex
- (4) 0.5, male

101. If the minimum weight and maximum weight of fruits produced by a plant is 50 gm and 200 gm respectively and this character is regulated by 3 pairs of genes. Then find out the weight of fruits produced by the progeny of the cross $AAbbCC \times aaBBcc$:-

- (1) 75 gm
- (2) 50 gm
- (3) 125 gm
- (4) 150 gm

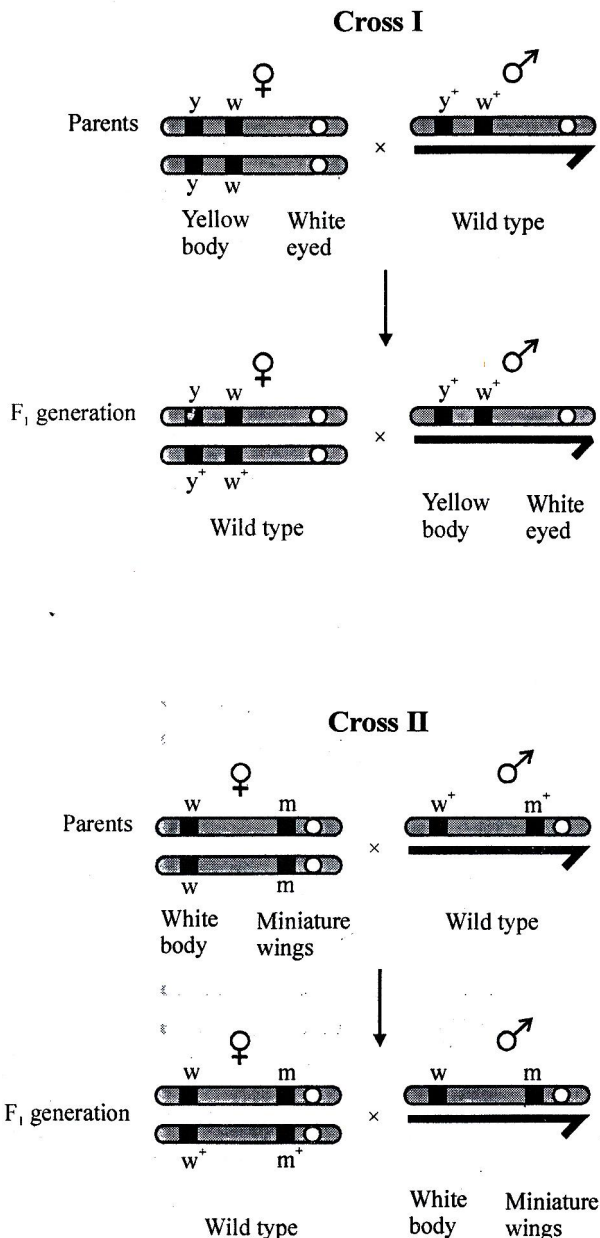
102. In an experiment, individual homozygous for 'ab' gene, were crossed with wild type. The F_1 hybrid was test crossed and progenies produced in following ratio :-
 $++/ab \rightarrow 896$
 $ab/ab \rightarrow 880$
 $+a/ab \rightarrow 108$
 $+b/ab \rightarrow 116$
 Calculate the distance between a and b gene
 (1) 22.4 cM (2) 44 cM
 (3) 11.2 cM (4) 45 cM
103. How many types of gametes are produced by a pentahybrid plant ($Tt Aa Bb Ee Gg$), if T and A are completely linked gene?
 (1) 32 (2) 16 (3) 8 (4) 2
104. The number of linkage group in human male and female are :-
 Male Female
 (1) 23 23
 (2) 23 24
 (3) 24 23
 (4) 22 23
105. The test cross of an F_1 individual produced the following offsprings-
 $++/ab - 10$
 $ab/ab - 10$
 $+a/ab - 40$
 $+b/ab - 40$
 Based on this data predict the configuration of gene :-
 (1) Cis
 (2) Trans
 (3) Both (1) & (2)
 (4) None of the above
106. In *Drosophila*, the allele for normal grey body colour (G) is dominant to ebony body (g). The following table summarises the results of several crosses:-
- | Cross | Result |
|----------------------|------------------|
| Strain 1 $\times$ gg | all Grey type |
| Strain 2 $\times$ gg | 1 Grey : 1 ebony |
| Strain 3 $\times$ gg | all ebony type |
| Strain 4 $\times$ Gg | 3 Grey : 1 ebony |
- Which strains have the genotype Gg ?
 (1) 1 and 3 (2) 1 and 4
 (3) 2 and 3 (4) 2 and 4

107. In a linear chromosome map, distance between 4 loci is as follow : a - b 10% ; a - d 3% ; b-c 4% and a-c 6%. The COV between c & d is
 (1) 3% (2) 9%
 (3) 3% or 9% (4) 4% or 12%
108. Possible blood group in children from the parents with 'B' and 'O' blood group are :
 (1) All B
 (2) All O
 (3) Both B and O
 (4) A and B
109. Bateson used term coupling and repulsion for linkage. Choose the correct coupling and repulsion combination :-
- | | Coupling | Repulsion |
|-----|------------|------------|
| (1) | AABB, aabb | AAbb, aaBB |
| (2) | AABB, aabb | AABB, AAbb |
| (3) | AAbb, aaBB | AaBb, aabb |
| (4) | aaBB, aabb | AABB, aabb |
110. Two genes A & B present on same chromosome show 20% recombination, then what will be the percentage of gamete (Ab) for the plant having genotype AaBb in cis configuration ?
 (1) 40 % (2) 80 %
 (3) 20 % (4) 10 %
111. Given below is, the figure of *Drosophila melanogaster*.



- It was found to be very suitable for experimental verification of chromosomal theory of inheritance by Morgan, because.
- (1) A single mating produces two young flies
 - (2) Smaller female is easily recognisable from larger male
 - (3) It completes life cycle in about two weeks
 - (4) It reproduces parthenogenetically

112. The experiment shown in the figure has been carried out by Morgan to show the phenomenon of linkage & recombination. If in Cross I, genes are tightly linked and in Cross II, genes are loosely linked then what will be percentage of recombinants produced in Cross I and Cross II respectively in F_2 generation ?



- (1) 98.7% and 62.8%
(2) 1.3% and 37.2%
(3) 37.2 and 1.3%
(4) 62.8% and 98.7%

113. Gene A and B are linked and are 18 map units apart. A heterozygous individual, whose parents were AAbb and aaBB, would be expected to produce gamete in the following frequencies :-
(1) 41%-AB, 9%-Ab, 9%-aB, 41%-ab
(2) 9%-AB, 41%-Ab, 41%-aB, 9%-ab
(3) 18%-AB, 32%-Ab, 32%-aB, 18%-ab
(4) 9%-AB, 9%-Ab, 41%-aB, 41%-ab
114. In which cross an organism showing a dominant phenotype and whose genotype is to be determined is crossed with the recessive parents:-
(1) Test cross
(2) Out cross
(3) Reciprocal cross
(4) Out cross with single character
115. In a Mendelian cross both genotypic and phenotypic ratio are same as 1 : 1 : 1 : 1. It represents a case of :-
(1) Monohybrid test cross
(2) Monohybrid cross with co-dominance
(3) Incomplete dominance
(4) Dihybrid test cross
116. A red flowered snapdragon plant is crossed with pink flowered plant, than 1000 plants are obtained how many of them are red flowered :-
(1) 500 (2) 250
(3) zero (4) 1000
117. Match the column-I with column-II

Column-I		Column-II	
I	Co-dominance	(A)	ABO blood group in human being
II	Multiple allele	(B)	Skin colour of human being
III	Incomplete dominance	(C)	Coat colour in cattle
IV	Polygenic inheritance	(D)	Size of starch grain in pea seeds

- (1) I-D, II-A, III-C, IV-B
(2) I-D, II-B, III-C, IV-A
(3) I-C, II-A, III-D, IV-B
(4) I-A, II-C, III-B, IV-D

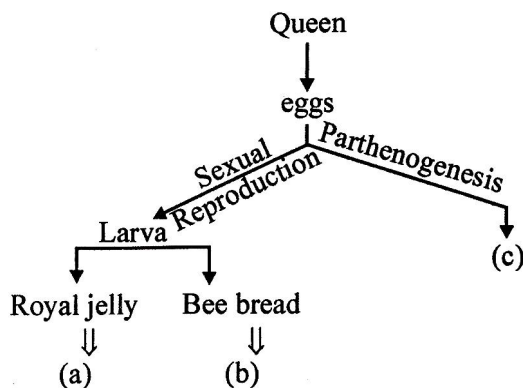
118. A normal female whose mother was colourblind married to a normal man whose father was colourblind, then find out percentage of their male child to be affected :-

- (1) zero (2) 50%
(3) 20% (4) 25%

119. Which of the following statement is incorrect :-

- (1) According to Mendel, characters are controlled by discrete units called factors.
(2) Fruit colour in summer squash is an example of dominant epistasis
(3) Haemophilia is an Autosomal linked, recessive disorder
(4) Recombination is used for generation of non-parental gene combination.

120. Identify the organism (a), (b) and (c) in the life cycle of honey bee shown below :



- (1) a = Queen ; b = Drone ; c = Worker
(2) a = Drone ; b = Worker ; c = Queen
(3) a = Worker ; b = Drone ; c = Queen
(4) a = Queen ; b = Worker ; c = Drone

121. Mendel obtained wrinkled seeds due to ?

- (1) Free sugar level is decreases
(2) Free sugar is more in place of starch
(3) Absence of starch branching enzyme (SBE)
(4) Presence of starch branching enzyme (SBE)

122. What will be phenotypes of F_1 generation by crossing variegated leaf colour female with pale or green colour male plant in mirabilis ?

- (1) Green
(2) Pale
(3) Variegated
(4) All of these

123. What will be phenotypic ratio of a dihybrid cross if one character is show complete dominance and another character is show incomplete dominance :-

- (1) 1 : 2 : 1 : 2 : 4 : 2 : 1 : 2 : 1
(2) 3 : 6 : 3 : 1 : 2 : 1
(3) 1 : 2 : 4 : 2 : 3 : 6 : 1
(4) 1 : 2 : 1

124. What is the percentage of parental types in a dihybrid cross between genes 'w' and 'm' performed by Morgan for showing strength of linkage in *Drosophila melanogaster* ?

- (1) 37.2%
(2) 62.8%
(3) 28.7%
(4) 10.3%

125. What is the percentage of parental phenotypes in F_2 generation for inheritance of skin colour in Human ?

- (1) 5.25%
(2) 2.25%
(3) 3.125%
(4) 6.125%

126. A tall red flowered plant was crossed with a dwarf white flowered plant. It produces heterozygous tall-red flowered offspring. On selfing of this dihybrid plant, it produces 16 offsprings. What is the ratio of offsprings which show parental phenotype :-

- (1) 1 : 1 : 1 : 1
(2) 9 : 3 : 3 : 1
(3) 9 : 1
(4) 10 : 2

127. A trihybrid AaBbCc plant produces :-

- (1) 8-types of gametes in ratio of
27 : 9 : 9 : 9 : 3 : 3 : 3 : 1
- (2) 8-types of gametes in ratio of
1 : 1 : 1 : 1 : 1 : 1 : 1 : 1
- (3) 27-types of gametes
- (4) None of the above

128. The term hemizygous means :-

- (1) A condition in which only one allele of a pair is present
- (2) Many alleles of each gene are present
- (3) Pair of contrasting traits are present
- (4) Two similar alleles are present

129. The recessive characters are :-

- (1) Only expressed in heterozygous condition
- (2) Only expressed in homozygous condition
- (3) Blended in heterozygous condition
- (4) Always impure

130. What is correct ?

- (1) Each back cross is test cross.
- (2) Each test cross is back cross.
- (3) Crossing F_2 with F_1 is called test cross.
- (4) Crossing F_2 with either parent is called test cross.

131. Types of gametes formed by the plant with genotype AABbccDD will be :-

- (1) 4
- (2) 16
- (3) 8
- (4) 2

132. In monohybrid cross, pure homozygous plants will be :-

- (1) one
- (2) two
- (3) three
- (4) four

133. Which of the following characters was not chosen by Mendel?

- (1) Pod shape
- (2) Pod colour
- (3) Location of flower
- (4) Location of seeds

134. Which of the following conclusion of Mendel is explained by dihybrid cross?

- (1) Dominance
- (2) Unit factor and segregation
- (3) Independent assortment
- (4) All of the above

135. Match the terms in Column-I with their description in Column-II and choose the correct option :

Column-I		Column-II	
(a)	Dominance	(i)	Many genes govern a single character
(b)	Co-dominance	(ii)	In a heterozygous organism only one allele expresses itself
(c)	Pleiotropy	(iii)	In a heterozygous organism both alleles express themselves fully
(d)	Polygenic inheritance	(iv)	A single gene influences many characters

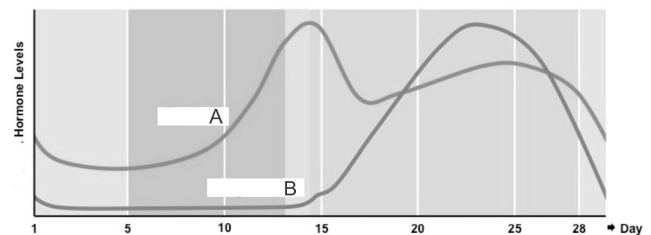
- | | | | |
|----------|-------|------|-------|
| (a) | (b) | (c) | (d) |
| (1) (ii) | (i) | (iv) | (iii) |
| (2) (ii) | (iii) | (iv) | (i) |
| (3) (iv) | (i) | (ii) | (iii) |
| (4) (iv) | (iii) | (i) | (ii) |

136. Absence of menstrual cycle is indicative of
(1) Pregnancy (2) Poor health
(3) Stress (4) All of above
137. Luteal phase of menstrual cycle is also known as
(1) Proliferative phase
(2) Secretory phase
(3) Follicular phase
(4) Ovulatory phase
138. Which of the following is not a function of epididymis ?
(1) Storage of spermatozoa.
(2) Transportation of spermatozoa.
(3) Maturation of spermatozoa.
(4) None of these
139. How many sperms and ova would be produced from 175 primary spermatocyte and 60 secondary oocyte
(1) 700 sperms and 240 ovum
(2) 240 sperms and 700 ovum
(3) 175 sperms and 240 ovum
(4) 700 sperms and 60 ovum
140. The germ cells (PGC) originates from
(1) yolk sac
(2) inside of gonads
(3) surface of gonads
(4) chorion
141. Process of sperm formation is under the regulatory influence
(1) Oxytocin and vitamin A
(2) FSH and vitamin E
(3) LH and Vitamin E
(4) Testosterone and vitamin C
142. A linear diagrammatic representation of Oogenesis is given below
- $$\begin{array}{ccc} A & \rightarrow & B \\ \downarrow & & \downarrow \\ \text{Primary} & & \text{Tertiary} \\ \text{folicle} & & \text{folicle} \end{array}$$
- What are the name of stage A and B of life :
- (1) A = Birth, B = Adult reproductive life
(2) A = Adult, B = Before birth
(3) A = B = Adult reproductive life
(4) A = IInd polar body , B = Ova.

143. What number of chromosome formed in fetal life during Oogenesis
(1) n (2) 2n (3) 4n (4) 3n
144. Fertilization occur in human and other placental mammals in -
(1) Ovary
(2) Uterus
(3) Ampullary- Isthmic Junction
(4) Oviduct.
145. The chemical nature of antifertilizin is -
(1) Acid protein (2) Glycoprotein
(3) Lipid (4) Carbohydrate
146. Oogenesis and spermatogenesis can be differentiated on the basis of
(1) Number of gametes produced by each primary gametocyte
(2) Number of gametes produced by each secondary gametocyte
(3) Yolk formation during growth phase
(4) All of above
147. Select the odd one on the basis of no. of chromosomes
(1) Primary spermatocyte
(2) Primary Oocyte
(3) Oogonia
(4) Secondary Oocyte
148. The spermatids are transformed into spermatozoa by the process, called
(1) Spermiogenesis
(2) Spermatocytosis
(3) Spermateliosis
(4) (1) and (3) both
149. Nucleus of sperm is
(1) Haploid, rounded, condense
(2) Diploid, rounded, condense
(3) Diploid, elongated, condense
(4) Haploid, elongated, condense
150. Select correct match
- | Column -I | Column-II |
|---------------------|--|
| Part of sperm | Function |
| (P) Acrosome | (A) Genetic information |
| (Q) Nucleus | (B) Energy for the movement |
| (R) Middle part | (C) Enzymes that help in fertilisation |
| (1) P -C, Q -A, R-B | (2) P -A, Q -C, R-B |
| (3) P -A, Q -B, R-C | (4) P -B, Q -A, R-C |

151. In human beings Oogenesis is completed
 (1) Before birth
 (2) Before puberty
 (3) After puberty
 (4) Between birth and puberty
152. In human beings ovulation is -
 (1) Release of secondary Oocyte at beginning of menstrual cycle
 (2) Release of ovum at mid of menstrual cycle
 (3) Release of ovum at 14th day of menstrual cycle
 (4) release of secondary oocyte at mid of menstrual cycle
153. What is the important role of acrosome in-
 (1) Sperm motility
 (2) Penetration of ovum by sperm
 (3) (1) & (2) both
 (4) hormone secretion
154. Number of chromosomes in zygote is
 (1) Equal to secondary oocyte
 (2) Equal to secondary spermatocyte
 (3) Equal to Ootid
 (4) Equal to primary spermatocyte
155. During Oogenesis second meiotic division is completed
 (1) In embryonal stage
 (2) Before puberty
 (3) Before ovulation
 (4) During fertilisation
156. Which of the following cells undergo meiosis-
 (1) Meiocyte of male
 (2) Meiocytes of female
 (3) Gamete mother cell
 (4) All of these
157. Which type of cell division occurs in gonads
 (1) Mitosis only
 (2) Meiosis only
 (3) Amitosis and Meiosis only
 (4) Both Mitosis and meiosis
158. Spermatogenesis is promoted by -
 (1) FSH (2) MSH
 (3) ACTH (4) hCG
159. In the male human being sperm contain one set of autosomes and -

- (1) Only one Y chromosome
 (2) Only one X chromosome
 (3) Both X and Y chromosome
 (4) Either X or Y chromosome
160. The products of the first maturation division of germ cell in testis are known as
 (1) Sperm
 (2) Oocytes
 (3) Spermatids
 (4) Secondary spermatocytes
161. Acrosome of sperm is formed from -
 (1) Nucleus of spermatid
 (2) Centrosome of spermatid
 (3) Mitochondria of spermatid
 (4) Golgi complex of spermatid
162. Identify the hormones labelled as A and B in the following graph.



- (1) A - LH, B - Progesterone
 (2) A - Oestrogen, B - FSH
 (3) A - Oestrogen, B - Progesterone
 (4) A - Progesterone, B - Oestrogen
163. How many polar bodies are formed during gametogenesis in males?
 (1) 2 (2) 1
 (3) 3 (4) none
164. In human sperm, Nebenkern refers to
 (1) Acrosome
 (2) Axial filament
 (3) Proximal centriole
 (4) Spiral Mitochondria
165. If in a sperm a distal centriole is removed, which of the following shall not occur?
 (1) Acrosomal reaction
 (2) Movement of sperm
 (3) First cleavage of zygote
 (4) Amphimixis

166. Find the odd statement.

- (1) Human male ejaculates about 200 - 300 million sperms during a coitus.
- (2) Type A spermatogonia serves as stem cell for spermatogenic lineage.
- (3) Spermiogenesis refers to maturation of sperms from their precursor spermatids.
- (4) Spermatids have 44 + XY chromosomes only.

167. Menarche is the stage when

- (1) First ejaculation occurs
- (2) First menstrual cycle occurs
- (3) Hymen breaks down
- (4) Parturition takes place

168. Stroma of the mammalian ovary consists of

- (1) Connective tissue
- (2) Nerves
- (3) Vascular tissue
- (4) All of these

169. Corpus luteum is developed from

- (1) Primary follicle
- (2) Ovarian stroma
- (3) Left over follicle after release of ovum
- (4) Ovum

170. Zona pellucida, a glycoprotein covering in mature follicle, is an example of

- (1) Primary egg membrane
- (2) Secondary egg membrane
- (3) Tertiary egg membrane
- (4) None of these

171. Which of the following statement is incorrect about menstruation?

- (1) Around 40 – 80 ml blood loss takes places.
- (2) Menses take place for 3 – 5 days.
- (3) Beginning of menstrual cycle at puberty is called menarche.
- (4) Menstrual fluid can clot easily.

172. The levels of which of the following hormone rises in urine, in females after menopause?

- (1) Oestrogen
- (2) Progesterone
- (3) FSH
- (4) Androgens

173. Estrous cycle is characteristic of

- (1) Human females
- (2) All vertebrates
- (3) All mammals
- (4) Non-primate mammalian females

174. Which of the following is also called 'yellow body'?

- (1) Corpus albicans
- (2) Corpus luteum
- (3) Graafian follicle
- (4) Cumulus oophorus

175. If the menstrual cycle of a woman is of 25 days, then ovulation occurs on

- (1) 11th day
- (2) 14th day
- (3) 20th day
- (4) 7th day

176. Mark the stage formed just after release of secondary oocyte (ovulation) due to rupturing of Graafian follicle.

- (1) Corpus luteum
- (2) Corpus albicans
- (3) Corpus epididymis
- (4) Corpus haemorrhagicum

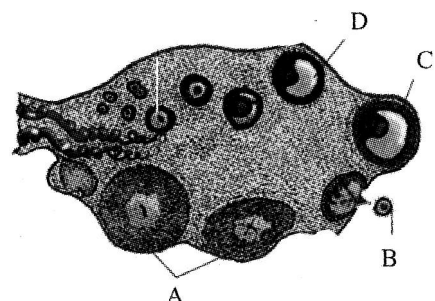
177. During ovulation in human female

- (1) Three peaks of oestrogen is present
- (2) Peak of FSH is higher than LH
- (3) Peak of LH is higher than FSH
- (4) Tallest peak of FSH is observed

178. Antrum, a fluid-filled cavity is typically seen in which follicle?

- (1) Primordial follicle
- (2) Primary follicle
- (3) Secondary follicle
- (4) Graafian follicle

179. The diagram of the sectional view of the ovary is given below. The labelled structure that secretes large amount of progesterone is



- (1) A
- (2) B
- (3) C
- (4) D

180. Second polar body is formed inside

- (1) Ovary
- (2) Fallopian Tube
- (3) Vagina
- (4) Uterus